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Academic Appointments

2024/Feb – present Postdoctoral associate in Chemistry, Massachusetts Institute of Technology, US

Education

2019/Sep – 2024/Jan Ph.D. in Chemistry, University of Massachusetts at Amherst, US
2015/Sep – 2019/Jul B.S. in Chemistry and Molecular Science, Wuhan University, China

Research Summary

Biomolecular Interactions: 1. specific binding in protein–protein interaction (PPI) network; 2. dynamic protein interactions among folded, intrinsically disordered, and multi-domain proteins, and post-translational modification (PTM) effects; and 3. liquid-liquid phase separation and membraneless biomolecular condensate formations.

Disease- and Health-Relevant Biophysical Mechanisms: mechanisms of dynamic protein signaling, regulatory or pathological processes: 1. bacterial immune evasion; 2. membrane pore transport and regulation; 3. neurodegeneration-associated protein phase separation; 4. nanopore-based single-molecule detection; 5. viral protease conformational dynamics and activity; and 6. oncogenic fusion proteins in cancer.

MD/ML-Integrated Biomolecular Modeling: 1. physics-based molecular dynamics, multiscale coarse-grained and atomistic force-field development, and theory-based off-lattice model development; 2. explainable and generative ML/AI methods for physical-model development and biomolecular analysis.

Bioinformatics: 1. protein language models (pLMs); 2. sequence–ensemble–function relationships; 3. evolutionary conservation and fitness landscapes; and 4. network biology.

Research Experience

02/2024 – : Postdoctoral Associate, Massachusetts Institute of Technology (Advisor: Bin Zhang)

5. MD/ML for predictive phase separation (2026/06 -).
I am developing a physics-grounded MD/ML method for quantitative prediction of disordered and multidomain protein phase separation.
4. Exploring the physical role of protein phase separation based on mean-field theory (2024/07 - 2026/07).
I developed the off-lattice based, sticker-and-random-spacer model to investigate how the balance between specific and non-specific interactions synergistically affects phase separation in homo- and heterogeneous systems. ([bioRxiv](#))
3. Developing a transferable coarse-grained protein model (2025/09 -2026/06).
I co-developed MOFF2, a machine-learning-assisted, physics-grounded coarse-grained force field that balances folded-domain stability, disordered-protein ensembles, multidomain organization, and intermolecular interactions for predictive condensate simulations. ([JCTC](#))
2. Resolving the earliest molecular events in biomolecular condensation (2024/07 – 2025/12).
In collaboration with Dr. Xin Zhang's experimental group, I integrated physics-based simulations with time-resolved measurements to characterize the initial microsecond of elastin-like protein condensation. The work connected early intermolecular association and conformational reorganization to subsequent condensate formation. ([JACS](#))
1. Decoding IDP functional roles using protein language model (2024/02 – 2024/10).
I led an explainable protein-language-model analysis connecting evolutionary fitness landscapes to IDP biophysics. We showed that phase-separating scaffold proteins are enriched in conserved disordered motifs whose residue identities follow sticker–spacer principles, providing a sequence-level framework for interpreting phase-separation function and disease-associated variation. ([eLife](#))

09/2019 – 02/2024: Graduate Research Assistant, University of Massachusetts Amherst (Advisor: Jianhan Chen)

1. Developing multi-scale enhanced sampling methods to study protein conformational dynamics.

1.3. Transferable atomistic protein model: C36mrb-disp (2023-2024)

With Dr. Gong, I co-developed [C36mrb-disp](#), a transferable atomistic model for both folded and disordered

proteins. We identified and corrected the overstabilized charge–charge interactions in CHARMM36m, improving disordered-protein ensembles while retaining folded-protein stability. ([JPCB](#))

1.2. Collective-variable independent enhanced sampling method: REST3 (2021-2022)

I developed REST3, a collective-variable independent enhanced sampling method that eliminates high-temperature conformational collapse and improves IDP sampling efficiency by approximately twofold. ([JCTC](#))

1.1. Physics-based protein coarse-grained model: HyRes II/HyRes-GPU (2021-2022)

I developed HyRes II, a transferable coarse-grained model with an explicit solvation term calibrated for long-range nonspecific interactions, enabling accurate simulations of IDP ensembles, dynamic protein interactions, and phase separation. I further optimized its GPU implementation to reach ~500 ns/day for systems with 100,000 beads. ([JCIM](#))

2. Application of multiscale simulation methods to studies of dynamic protein interactions.

2.4. IDP phase separation: how local structure affects the phase separation (2023-2024)

With coarse-grained simulations, I showed that phase behavior in GY-23 and TDP-43 is governed by the balance of backbone- and side-chain-mediated interactions rather than secondary-structure propensity alone, recapitulating experimental and mutation-dependent trends. ([JACS](#))

2.3. Real-time biomolecule sensing: nanopore engineering for tracking flavivirus proteases functional conformations (2021-2023)

With Dr. Min Chen's lab, I combined molecular simulations and electrophysiology to engineer a ClyA nanopore that captures flaviviral proteases and distinguishes functional conformational states at the single-molecule level. ([Biophys. J.](#))

2.2. Bacterial infection: immune evasion of staphylococcus (2020-2021)

Inspired by NMR data from Dr. Brian V. Geisbrecht's studies, I investigated how the conformational dynamics of Staphylococcal peroxidase inhibitor (SPIN) help the bacteria evade human immune responses. ([Frontiers](#))

2.1. Regulation of mitochondrial pore opening: interactions between p53-TAD/CypD (2020-2021)

Working with experimental collaborators, I used simulations to reveal how disordered regions regulate molecular recognition: p53-TAD binds cyclophilin D through a dynamic electrostatic interface that promotes mitochondrial pore opening. The predicted mechanisms were supported by NMR, SPR, and biochemical measurements. ([JMB](#))

07/2018-05/2019: Undergraduate Researcher, Wuhan University, China (Advisor: Fuan Wang)

1. Hybrid HCR-CHA circuit for phosphorylase kinase signal amplification (2018-2019)

We developed an HCR-CHA circuits to amplify the FRET signal during a continuous chain reaction, which can sensitively and specifically detect the signals of phosphorylase kinase at nanomolar concentrations.

Collaboration Experience

Nuclear magnetic resonance (NMR) Labs:

- Dr. Mei Hong's lab at Massachusetts Institute of Technology: Tau protein fuzzy coat functions.
- Dr. Brian V. Geisbrecht's lab at Kansas State University: Immune inhibition mechanism by bacteria.
- Dr. Chunyu Wang's lab at Rensselaer Polytechnic Institute: PPI regulation for mitochondrial pore opening.

Nanopore Technology Labs:

- Dr. Min Chen's lab at UMass Amherst: Nanopore tweezer engineering for enzyme functional states detecting.

Infrared (IR) spectroscopy Labs:

- Dr. Xin Zhang's lab at West Lake University: Early-stage condensate assembly.

Future Research Interests

1. Explainable MD/ML model for biomolecular functional investigation and prediction.
2. Biophysics in human diseases including immunology, cancer, and neurodegeneration.
3. Network biology including protein-protein interactions, gene regulatory networks, and signaling networks.
4. Bioinformatics for biomolecular function and evolutionary analysis.
5. Predictive and generative model for peptides/small molecules design targeting pathological biological activities.

Publications (#: co-first authors; *: corresponding authors ; [†]: **key publications**)At MIT:

[15]. **Y. Zhang**[#], A. Sood[#], A. Athreya, B. Zhang^{*}, "Valency-Limited Molecular Dynamics Simulations of Stickers-and-Spacers Polymers Reveal a Tradeoff Between Condensation and Organization." (2026) (under minor revision) [bioRxiv](#)

[14]†. S. Liu[#], **Y. Zhang**[#], I. Riveros, C. Wang, B. Zhang^{*}, "MOFF2: A Transferable Coarse-Grained Protein Force Field for Predictive Condensate Simulations." *J. Chem. Theory Comput.* (2026). [ACS](#)

[13]†. L. Zhu[#], G. Liao[#], **Y. Zhang**[#], Y. Guo, J. Chen, S. Zhou, Z. Wu, Y. Huang, X. Lu, C. Chen, H. Chen, Y. Weng^{*}, X. Zhang^{*}, "A Glimpse into the Initial Microsecond of Biomolecular Condensation." *J. Am. Chem. Soc.* (2026). [ACS](#)

[12]†. **Y. Zhang**, J. Zheng, B. Zhang^{*}, "Protein Language Model Identifies Disordered, Conserved Motifs Implicated in Phase Separation." *eLife*. (2025). [eLife](#)

At UMass Amherst:

[11]. **Y. Zhang**[#], X. Liu[#] and J. Chen^{*}, "Intrinsically disordered proteins", in "Generalized-Ensemble Algorithms - Ideas and Applications", Edited by Sugita and Okamoto, Springer, 2026. (*In Press, Submitted to Springer*)

[10]. S. Barethiya[#], S. Schultz[#], **Y. Zhang**, J. Chen^{*}, "Coarse-Grained Simulations of Phosphorylation Regulation of p53 Autoinhibition." *Biochemistry*. (2025). [ACS](#)

[9]. S. A. Shorkey[#], **Y. Zhang**[#], J. Sharp[#], S. Clingman, L. Nguyen, J. Chen^{*} and M. Chen^{*}, "Tracking flaviviral protease conformational dynamics by tuning single-molecule nanopore tweezers.", *Biophys J.* (2025). [ScienceDirect](#)

[8]. X. Gong[#], **Y. Zhang**[#], J. Chen. "Likely Over-Stabilization of Charge-Charge Interactions in CHARMM36m(w): A Case for a99SB-disp Water.", *J. Phys. Chem. B.* (2024). [ACS](#)

[7]. S. Li, **Y. Zhang** and J. Chen^{*}, "Backbone Interactions and Secondary Structures in Phase Separation of Disordered Proteins.", *Biochem. Soc. Trans.* (2024). [Portland Press](#)

[6]†. **Y. Zhang**[#], S. Li[#], X. Gong and J. Chen^{*}, "Toward Accurate Simulation of Coupling between Protein Secondary Structure and Phase Separation.", *J. Am. Chem. Soc.* (2024). [ACS](#)

[5]†. **Y. Zhang**, X. Liu^{*} and J. Chen^{*}, "Re-balancing replica exchange with solute tempering for sampling dynamic protein conformations", *J. Chem. Theory Comput.* 19, 1602 (2023). [ACS](#)

[4]. **Y. Zhang**[#], X. Liu[#] and J. Chen^{*}, "Coupled binding and folding of SPIN N-terminal region in myeloperoxidase inhibition", *Front. Mol. Biosci.* (2023). [frontiers](#)

[3]†. **Y. Zhang**, X. Liu^{*} and J. Chen^{*}, "Towards Accurate Coarse-Grained Simulations of Disordered Proteins and Their Dynamic Interactions", *J. Chem. Inf. Model.* 62, 4523-4536 (2022) [ACS](#)

[2]. J. Zhao, X. Liu, A. Blayney, **Y. Zhang**, L. Gandy, F. Zhang, R. J. Linhardt, J. Chen, C. Baines, S. N. Loh and C. Wang, "Intrinsically disordered N-terminal domain (NTD) of p53 interacts with mitochondrial PTP regulator Cyclophilin D" *J. Mol. Biol.* 434, 167552 (2022). [JMB](#)

[1]. X. Gong[#], **Y. Zhang**[#] and J. Chen, "Advanced Sampling Methods for Multiscale Simulation of Disordered Proteins and Dynamic Interactions" *Biomolecules*, 11, 1416 (2021) (Invited Review). [MDPI](#)

Talks and Poster Presentation Experience

Talks:

- 2023, Chemistry ResearchFest, UMass Amherst
- 2022, Chemistry-Biology Interface (CBI) Chalk Talk, UMass Amherst

Posters:

- 2024, Biophysics Retreat, MIT.
- 2023, American Chemical Society Spring
- 2022, Biophysical Society Annual Meeting
- 2020 - 2022, Chemistry ResearchFest, UMass Amherst

Teaching and Mentoring Experience

Teaching (as teaching assistant):

- General Chemistry (Chem 111 & Chem 112), UMass Amherst
- Advanced Physical Chemistry (Chem 585), UMass Amherst

Mentoring:

- Graduate students: Samantha Schultz, Juni Campbell, Shrishti Barethiya and Kairong Dong.
- Undergraduate students: Anik Dey, Ryan Pham, Jiayue Wang, Sasha Zhang.

Technical Skills

Programming Skills: Highly proficient in Linux operating systems, Python, and PyTorch

Computational and Molecular Modeling: Extensive experience in biological system modeling and analyzing. Rich experience in multi-scale state-of-the-art force fields such as CHARMM, Amber, HPS, and Martini. Expertise in advanced sampling methods, including Umbrella Sampling, Replica Exchange, Free Energy Perturbation, and Thermodynamic Integration.

Data Analysis: Highly skilled in data curation and large-scale data analysis, including deep learning-based clustering, principal component analysis (PCA), factor analysis, and feature extraction techniques.

Machine Learning: Proficient in large language model, contrastive learning, diffusion model, score/flow-matching

Funded Research Activities

2025–present: National Synthesis Center for Emergence in the Molecular and Cellular Sciences (NCEMS; NSF-funded), Working Group Member, [“Ocean of Disorder”](#), Pennsylvania State University.

Professional Service

2021–present Peer Reviewer for:

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| • <i>Biophysical Journal</i> | • <i>Physical Chemistry Chemical Physics</i> |
| • <i>Scientific Reports</i> | • <i>PLOS Computational Biology</i> |
| • <i>Proteins</i> | • <i>Biochemical and Biophysical Research Communications</i> |
| • <i>Journal of Chemical Information and Modeling</i> | • <i>Journal of Physical Chemistry B</i> |
| • <i>Journal of Chemical Theory and Computation</i> | • <i>ACS Omega</i> |
| • <i>International Journal of Biological Macromolecules</i> | • <i>Journal of the American Chemical Society Au</i> |

Honors and Awards

- 2023** Dr. Paul Hatheway Terry Endowment Award in Chemistry ResearchFest, UMass Amherst
- 2020** 1st place prize in Chemistry ResearchFest poster presentation, UMass Amherst
- 2016** 2nd Class Freshman Scholarship, Wuhan University, China
- 2015** Freshman Scholarship, Wuhan University, China